

Network Intrusion Detection Using CIC-IDS-2017

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1. Introduction

This project investigates network intrusion detection using the CIC-IDS-2017 dataset published by the Canadian Institute for Cybersecurity. Cyberattacks pose a growing threat to modern infrastructure, and the ability to automatically classify malicious network traffic is a critical challenge in cybersecurity. The goal of this analysis is to determine whether network flow features can be used to accurately identify and classify different types of cyberattacks using machine learning models in R.

2. Dataset Description

The CIC-IDS-2017 dataset consists of eight CSV files capturing network traffic recorded over five days (Monday through Friday) in a controlled lab environment. After combining all files and removing rows with infinite or missing values, the final cleaned dataset contains 2,827,876 observations and 79 variables. The 78 feature variables are all quantitative and describe network flow statistics such as packet lengths, flow duration, inter-arrival times, byte rates, and TCP flag counts. The target variable, Label, is categorical and identifies each flow as either benign or one of 13 attack types including DDoS, PortScan, DoS Hulk, FTP-Patator, SSH-Patator, Bot, and Web Attacks. An additional implicit categorical variable is the time period (day of week) from which each file originates.

3. Research Questions

This analysis investigates the following questions. First, can network flow features accurately classify traffic as benign or as a specific attack type? Second, which features are the most important predictors of attack type? Third, how does class imbalance affect model performance, and which model handles minority attack classes better — Random Forest or XGBoost?

4. Descriptive Statistics

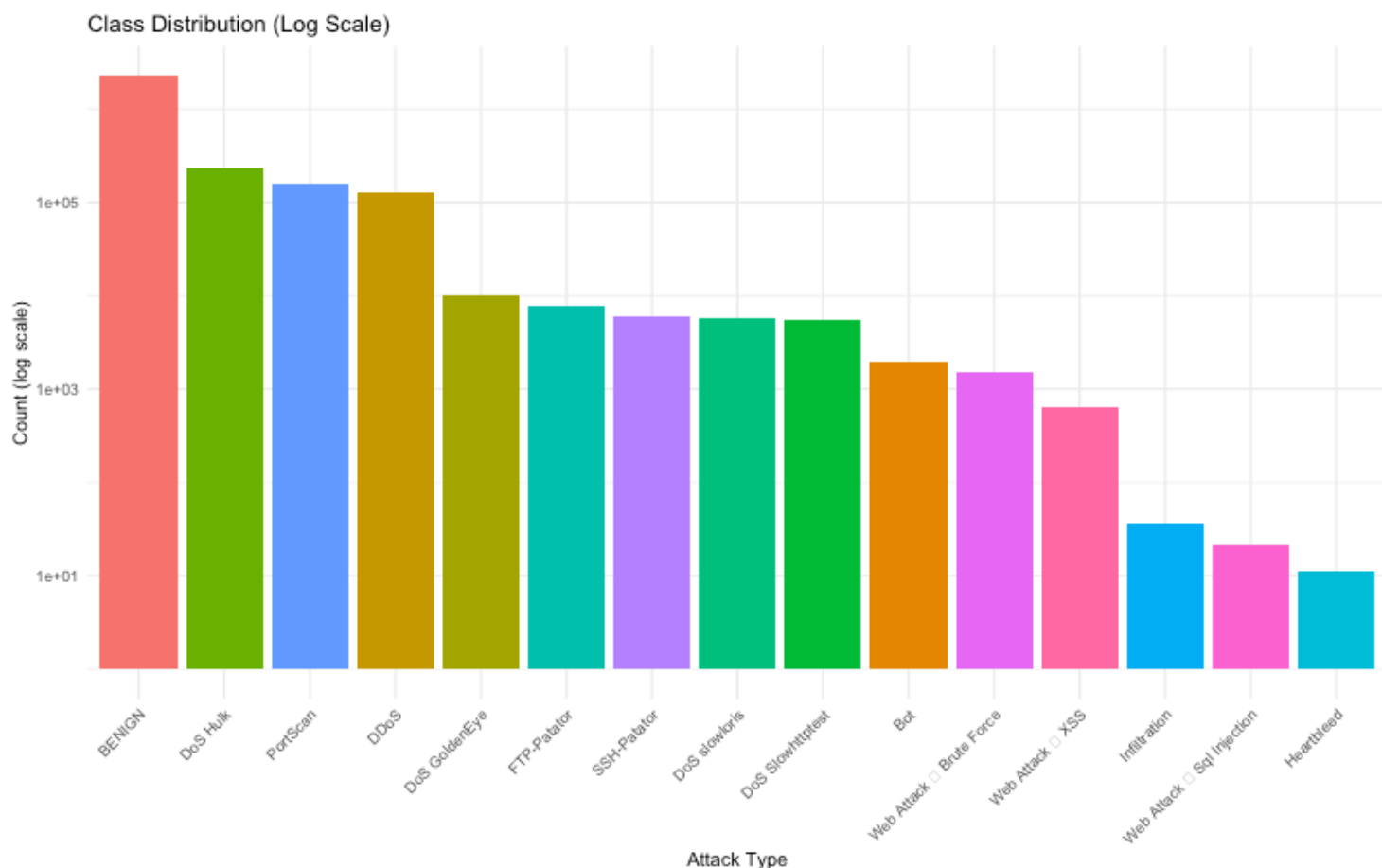
The table below summarizes key quantitative features across the dataset. Flow Bytes per second shows an extremely large range (from negative values caused by zero-duration flows up to 2 billion), reflecting the diverse nature of network traffic. Packet Length Standard Deviation and Average Packet Size similarly show high variance, suggesting that attack traffic differs substantially from normal traffic in its packet structure. Flow IAT Mean (inter-arrival time) ranges from near zero to 120 million microseconds, capturing both high-speed attack floods and slow-rate attacks.

	Feature	Mean	Median	SD	Min	Max
1	Flow.Bytes.s	1443681.82	4557.22	23550597.90	-1.2e+07	2.07100e+09
2	Packet.Length.Std	294.02	25.86	629.43	0.0e+00	4.73152e+03
3	Average.Packet.Size	191.50	72.25	330.54	0.0e+00	3.33714e+03
4	Fwd.Packet.Length.Mean	58.24	34.00	185.53	0.0e+00	5.94086e+03
5	Bwd.Packet.Length.Std	334.10	0.00	836.22	0.0e+00	6.71574e+03
6	Flow.IAT.Mean	1314840.66	11528.43	4585215.10	-1.0e+00	1.20000e+08

Descriptive Statistics

5. Class Distribution

The dataset is severely imbalanced, as shown in the figure below. Benign traffic accounts for over 2.2 million records, while some attack types are extremely rare — Infiltration has only 36 samples and SQL Injection has only 21 samples across the entire dataset. This imbalance is a fundamental challenge for classification, as models trained on such data tend to favor the majority class. A log scale is used on the y-axis to make minority classes visible.



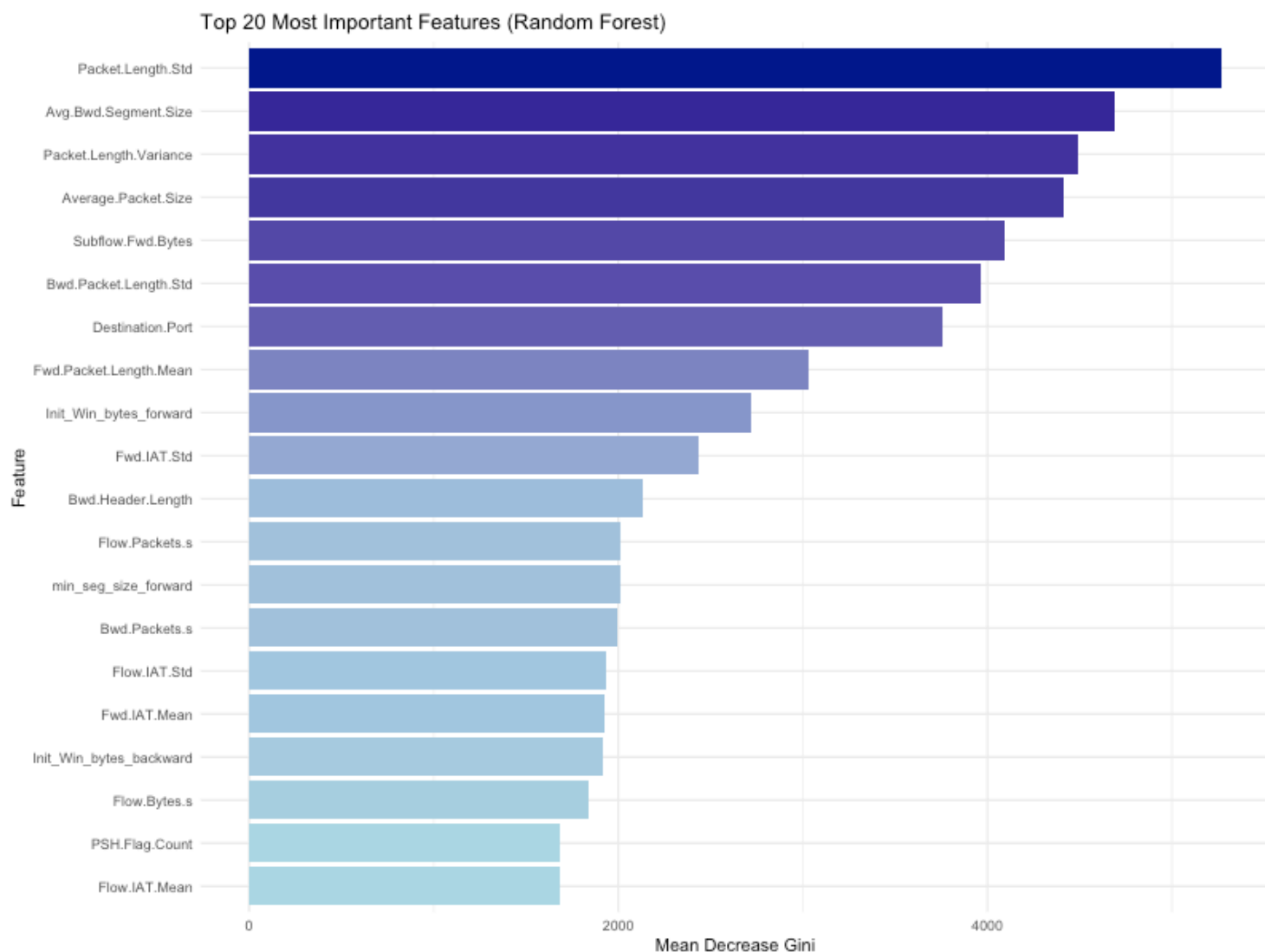
Class Distribution

6. Data Preprocessing

Several preprocessing steps were applied before modeling. Columns with zero variance were removed as they carry no predictive information. Features with pairwise correlation above 0.95 were also removed to reduce redundancy, as shown in the correlation heatmap. Min-max normalization was applied to all remaining numeric features to place them on a common scale. A stratified 10% sample of 282,781 rows was drawn to make model training computationally feasible, and classes with fewer than three samples (Heartbleed and SQL Injection) were excluded from modeling. The data was then split into 80% training (226,230 rows) and 20% testing (56,550 rows) using stratified partitioning.

7. Feature Importance

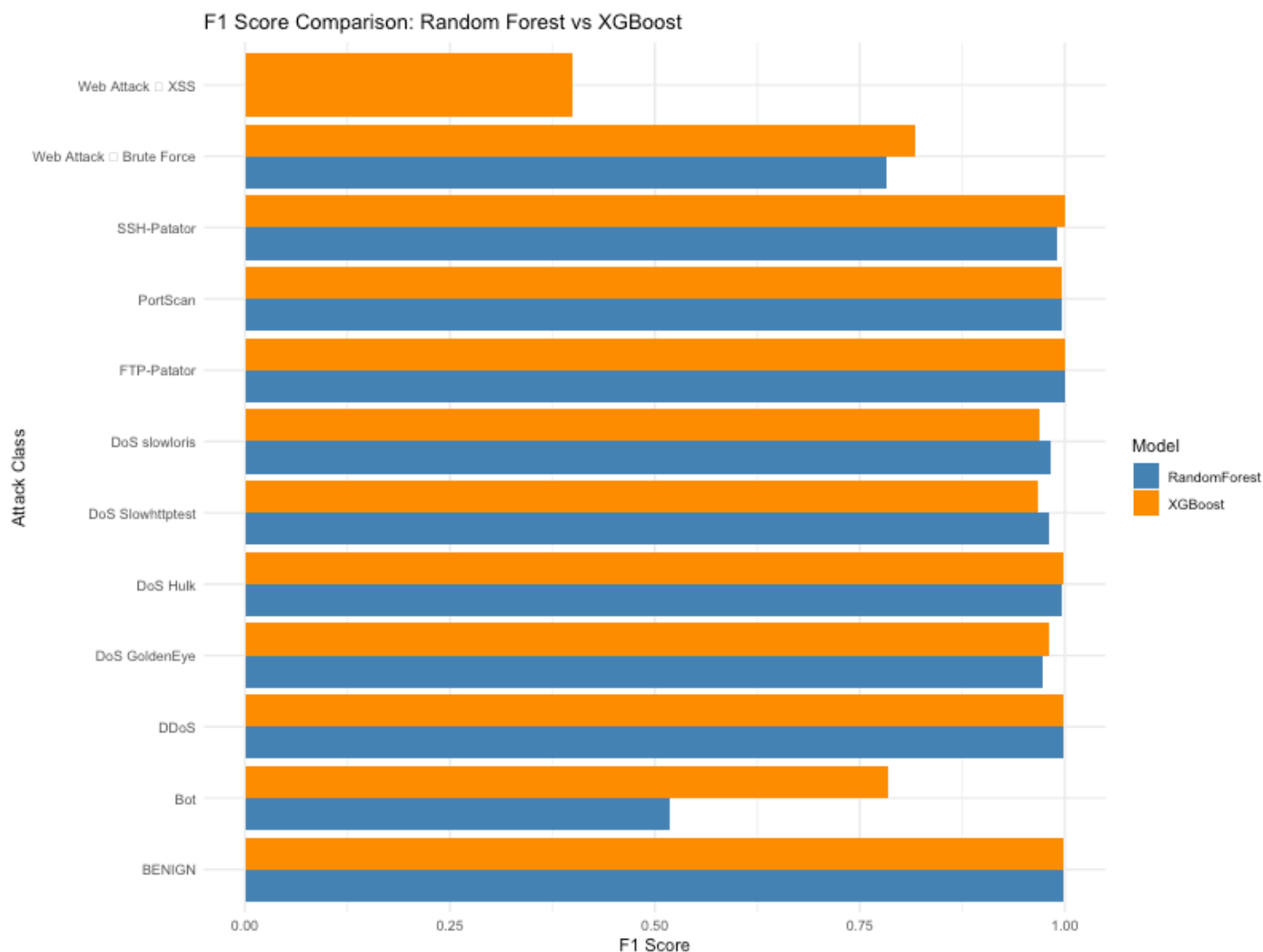
After training the Random Forest model, feature importance was assessed using the Mean Decrease in Gini impurity. The figure below shows the top 20 most important predictors. Packet Length Standard Deviation, Average Backward Segment Size, and Packet Length Variance are the three most discriminative features. This suggests that attacks are most distinguishable from benign traffic by the variability and size of their packets rather than their timing or port usage. Destination Port, while ranked 7th, also plays a meaningful role, consistent with the fact that certain attacks target specific ports.



Feature Importance

8. Model Results

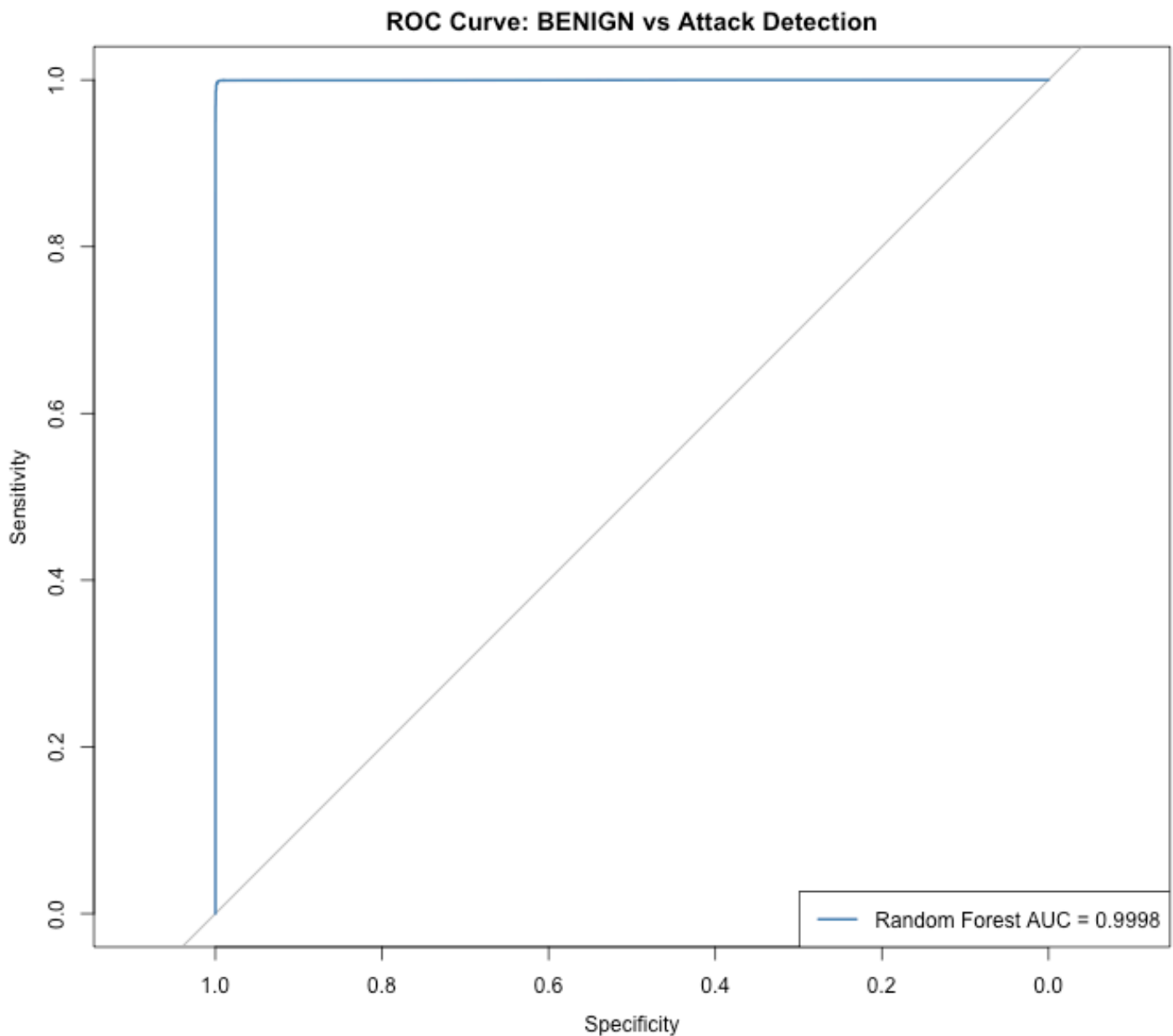
Two classification models were trained and compared: Random Forest with 500 trees and XGBoost with 100 boosting rounds. The Random Forest model achieved an overall out-of-bag error rate of 0.2%, indicating excellent overall performance. On the test set, both models achieved near-perfect F1 scores for well-represented classes such as BENIGN (0.999), DDoS (0.999), FTP-Patator (1.000), and PortScan (0.996). However, performance differed notably on minority classes. XGBoost achieved an F1 score of 0.75 for Bot traffic compared to 0.52 for Random Forest, demonstrating that gradient boosting handles class imbalance better in this context. Classes with very few samples such as Infiltration and Web Attack XSS could not be reliably detected by either model, yielding NA or near-zero F1 scores.



F1 Score Comparison

9. ROC Curve Analysis

The ROC curve below evaluates the Random Forest model ability to distinguish benign traffic from any attack type. The model achieved an AUC of 0.9998, indicating near-perfect discrimination. The curve rises sharply to the top-left corner, confirming that the model can detect attacks with very high sensitivity while maintaining an extremely low false positive rate. This is particularly important in real network environments where falsely flagging benign traffic would disrupt normal operations.



ROC Curve

10. Conclusion

This analysis demonstrates that machine learning models trained on network flow statistics can classify cyberattacks with high accuracy. Both Random Forest and XGBoost achieved near-perfect performance on well-represented attack types, with overall error rates below 0.2%. XGBoost outperformed Random Forest on minority attack classes, making it the preferred model for real-world deployment where rare attacks are most dangerous. The primary limitation of this analysis is severe class imbalance — attack types with very few samples could not be reliably detected. Future work should explore oversampling techniques such as SMOTE to improve minority class detection. The most important predictors were packet length statistics, confirming that attacks generate distinctive packet size patterns that distinguish them from normal traffic.